

Assignment8

2025-10-29

```
install.packages("readxl")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(readxl)
install.packages("ggplot2")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(ggplot2)
install.packages("dplyr")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

install.packages("GGally")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library("GGally")
install.packages("tidyverse")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.1      v stringr 1.5.2
## v lubridate 1.9.4    v tibble 3.3.0
## v purrr 1.2.0       v tidyr 1.3.1
## v readr 2.1.6

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()      masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

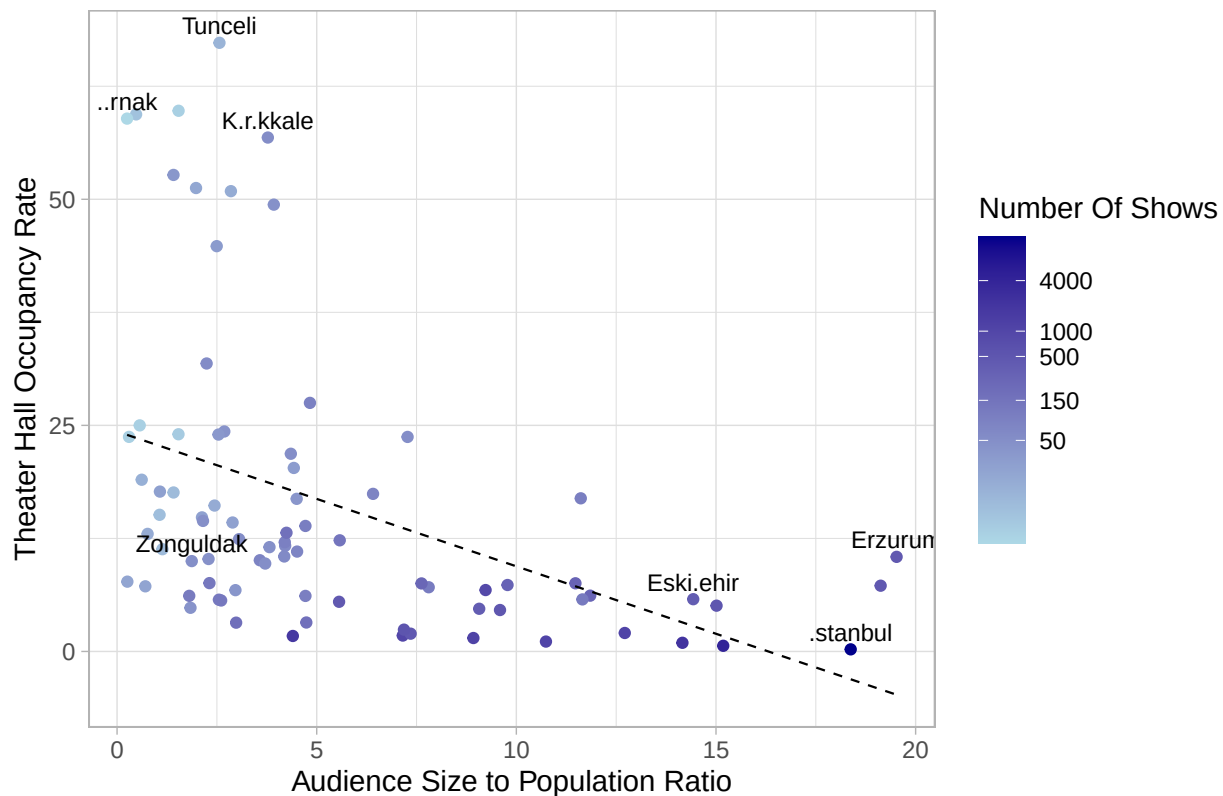
data<-read_excel("theatre_data.xlsx")

etiket<- data %>%
  filter(citys %in% c("İstanbul","Zonguldak","Şırnak","Tunceli","Eskişehir","Kırıkkale","Erzurum"))

ggplot(data, aes(x = ap_rate, y = s_rate, colour = perf)) +
  geom_point() +
  geom_text(data=etiket,aes(label=citys),
            vjust=-0.5,
            hjust=0.5,
            size=3,
            color="black")+
  geom_smooth(method=lm , color="black", se=FALSE, linetype="dashed",linewidth=0.4)+
  labs(title = "Correlation Between Theater Attendance-Hall Occupancy Rate By Province",
       x="Audience Size to Population Ratio",
       y="Theater Hall Occupancy Rate",
       colour="Number Of Shows")+
  scale_color_gradient(low="lightblue",high="darkblue",trans="log10",breaks=c(50,150,500,1000,4000))+
  guides(color=guide_colorbar(barheight=8))+
  theme_light()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Correlation Between Theater Attendance-Hall Occupancy Rate By Province



As theater attendance increases in cities relative to their population, theater halls occupancy rates decrease. In other words, there's a negative correlation between them. In cities like Istanbul, Erzurum, and Eskişehir, where the majority of the population attends theaters, theater occupancy rates are low. Conversely, in cities

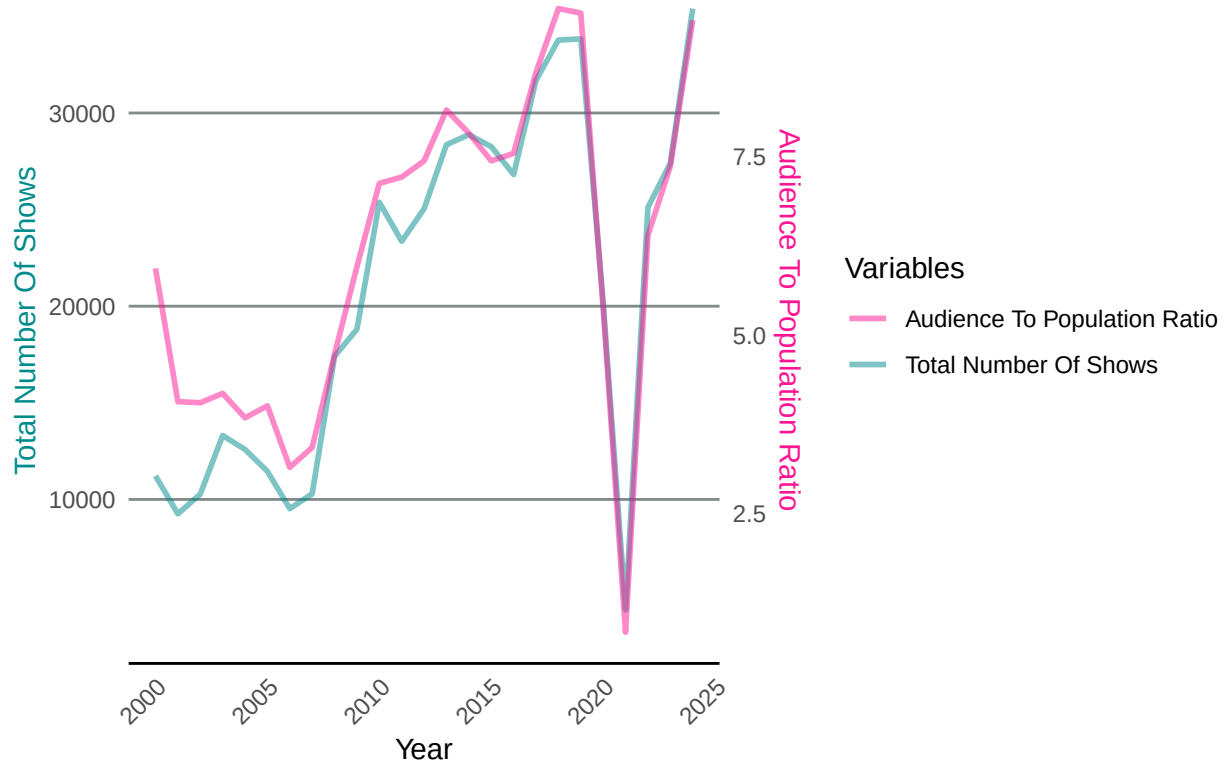
like Şırnak, Tuncali, and Kırıkkale, where theater attendance is low relative to the population, theater halls occupancy rates are higher. This negative correlation stems from the fact that cities with high theater attendance rates have higher performance numbers. In cities with high attendance rates, the number of performances and greater variety of performances result in lower theater occupancy rates.

```
k <- max(data$y_perf, na.rm = TRUE) / max(data$y_ap_rate, na.rm = TRUE)
ggplot(data, aes(x = year)) +
  geom_line(aes(y = y_perf, color = "Total Number Of Shows"), linewidth = 1,alpha=0.5) +
  geom_line(aes(y = y_ap_rate * k, color = "Audience To Population Ratio"), linewidth = 1,alpha=0.5) +
  scale_y_continuous(
    name = "Total Number Of Shows",
    sec.axis = sec_axis(~./k, name = "Audience To Population Ratio")) +
  scale_color_manual(name = "Variables", values = c("Total Number Of Shows" = "darkcyan", "Audience To Population Ratio" = "deeppink")) +
  labs(title = "Time Series Comparison",
       x="Year") +
  theme_minimal() +
  theme(
    panel.border = element_blank(),
    panel.grid.major.x = element_blank(),
    panel.grid.minor.x = element_blank(),
    panel.grid.major.y = element_line(color = "azure4", linetype = "solid",size=0.5),
    panel.grid.minor.y = element_blank(),
    axis.line.x = element_line(color = "black"),
    axis.line.y = element_blank(),
    axis.title.y.left = element_text(color = "darkcyan"),
    axis.title.y.right = element_text(color = "deeppink"),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## Warning: Removed 56 rows containing missing values or values outside the scale range
## (`geom_line()`).
## Removed 56 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

Time Series Comparison



This time-series graph illustrates the increasing interest in theater over the years and how Türkiye has responded. While theater attendance was high in 2000, this rate generally declined over the years, with occasional slight increases, until approximately 2008. After this decline, the number of performances began to increase as attendance began to rise. Generally, the graph shows that as attendance increased, the number of performances increased, and as attendance decreased, the number of performances decreased. After 2020, due to the decrease in attendance due to the coronavirus, the number of performances also decreased, and this began to increase after the pandemic ended. Consequently, Turkey increased the number of performances while attendance increased. In other words, we can say that the theater sector is being prioritized.

```
install.packages("devtools")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
library(devtools)
```

```
## Loading required package: usethis
```

```
devtools::install_github("htastan/TRmaps")
```

```
## Skipping install of 'TRmaps' from a github remote, the SHA1 (c136d9fa) has not changed since last install
## Use `force = TRUE` to force installation
```

```
library(TRmaps)
```

```
install.packages("sf")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
library(sf)
```

```
## Linking to GEOS 3.8.0, GDAL 3.0.4, PROJ 6.3.1; sf_use_s2() is TRUE
```

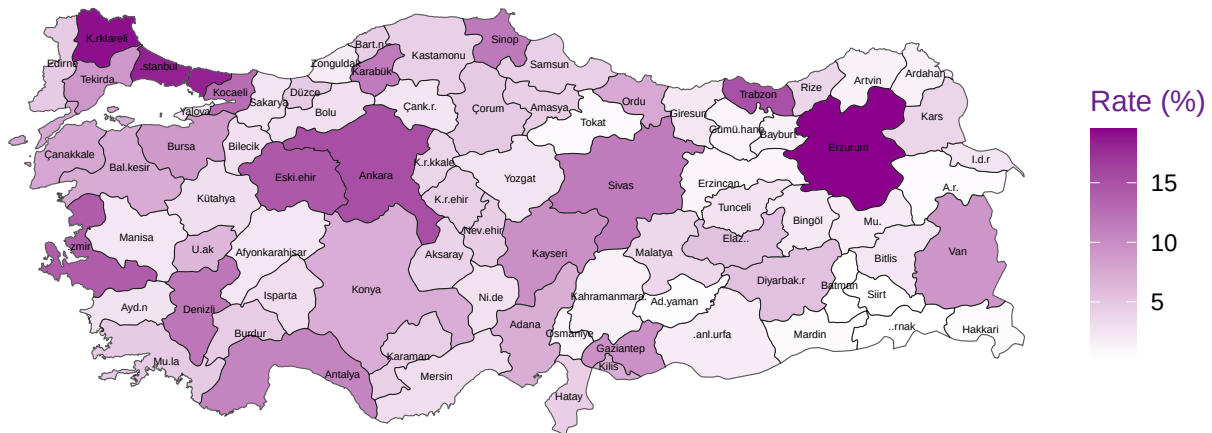
```
data("tr_nuts3")
```

```
data2 <- left_join(tr_nuts3, data, by = c("name_tr" = "citys"))
```

```
ggplot(data2) +
  geom_sf(aes(fill = ap_rate)) +
  geom_sf_text(aes(label=name_tr),size=1.5)+
  theme_void()+
  theme( plot.title= element_text(
    color="darkorchid4" , face="bold" ),
    legend.title= element_text(
    color="darkorchid4"
  ))+
  scale_fill_gradient(low = "white",high="darkmagenta")+
  labs(title="Audience Size to Population Ratio in 2023/2024 Season",
    fill="Rate (%)")
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```

Audience Size to Population Ratio in 2023/2024 Season



In cities like Erzurum, Istanbul, Kırklareli, and Ankara, shown in darker colors on the map, the majority of the population attends the theater. In other words, interest in theater is high in these cities. In cities with lighter colors, the proportion of theatergoers relative to the population is low. In other words, as the color tone on the map lightens, interest in theater decreases.